SymPy Bug Card

Bug 18: sinh incorrectly evaluates $I\pi + I(\pi - 1)$ to zero

Task. Evaluate a hyperbolic sine at a split imaginary-period argument.

$$\sinh(I\pi + I(\pi - 1))$$

SymPy's answer.

$$\sinh(I\pi + I(\pi - 1)) = 0.$$

Correct answer.

$$\sinh(I\pi + I(\pi - 1)) = -I\sin(1).$$

Explanation. The argument satisfies $I\pi + I(\pi - 1) = I(2\pi - 1)$, and $\sinh(It) = I\sin(t)$. Hence the value is $I\sin(2\pi - 1) = -I\sin(1)$, not zero.

Likely root cause. The diagnosis identifies `_peeloff_pi` in `sympy/functions/elementary/trigonometric.py` as peeling the π-coefficient from the nested addend $\pi - 1$ while dropping the non-π remainder. The deduplication assessment classifies this as a new instance of a known failure mode.

Artifact (GitHub).
bug-report/bug-018-sinh-incorrectly-evaluates-i-pi-i-pi-1-to-zero/